Flow of episode-level records to the five pathogen-specific modelling sets

Consolidated episode-level GEMS case–control dataset

n = 36,809 episode-level records

GEMS1: 22,567 + GEMS1A: 14,242

Core climate-variable availability

available for all 36,809 records

extracted from the ERA5-Land reanalysis for each field site; no records excluded at this stage

Pathogen-specific outcome-label availability and analytic-set construction

applied separately to each pathogen; both stages are shown for every branch

Shigella	ST-ETEC	Rotavirus	Cryptosporidium	Norovirus
Outcome-label availability **n = 36,809** available	Outcome label unavailable **n = 0**	Outcome-label availability **n = 36,809** available	Outcome label unavailable **n = 0**	Outcome-label availability **n = 11,325** available
Site-level contextual demographic covariate required to be non-missing	Excluded **n = 10,424**	Site-level contextual demographic covariate required to be non-missing	Excluded **n = 10,424**	Site-level contextual demographic covariate required to be non-missing
Final pathogen-specific modelling set **n = 26,385** GEMS1 13,416 + GEMS1A 12,969 7 countries, 40 SHCs		**Final pathogen-specific modelling set** **n = 26,385** GEMS1 13,416 + GEMS1A 12,969 7 countries, 40 SHCs		**Final pathogen-specific modelling set** **n = 6,111** GEMS1 records only 7 countries, 38 SHCs

Cryptosporidium	Norovirus
Outcome label unavailable **n = 25,484**	Outcome-label availability **n = 11,334** available
Site-level contextual demographic covariate required to be non-missing	Excluded **n = 5,228**
Final pathogen-specific modelling set **n = 6,106** GEMS1 records only 7 countries, 38 SHCs	

Norovirus	
Outcome label unavailable **n = 25,475**	Outcome-label availability **n = 11,329** available
Site-level contextual demographic covariate required to be non-missing	Excluded **n = 5,218**
Final pathogen-specific modelling set **n = 6,111** GEMS1 records only 7 countries, 38 SHCs	

The analysis master was restricted at construction to the case-control recruitment family (GEMS1 + GEMS1A); no health-utilisation-survey records were present.
Shigella and ST-ETEC were modelled on the same 26,385 records; the rotavirus, Cryptosporidium, and norovirus modelling sets contained GEMS1 records only.
Counts are episode-level records. Pathogen-specific sets are not mutually exclusive and must not be summed across branches.
The definition of the site-level contextual demographic covariate, and the coding of negative, invalid, and untested results, are given in Methods.
All five modelling sets span the same seven countries.